

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

MAY EXAMINATION

AMCS2104 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

THURSDAY, 7 MAY 2026

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

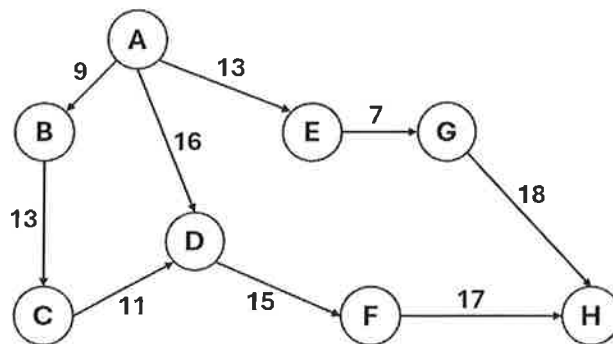
AMCS2104 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCEQuestion 1

- a) An *optimisation problem* is the task of finding the best solution from all feasible solutions. Provide **TWO (2)** examples of *optimisation problem*. (2 marks)
- b) Unlike path-searching problems that are usually addressed using *classical search algorithms*, *optimisation problems* can often be solved using *local search algorithms*. Differentiate between *local search algorithms* and *classical search algorithms* in **TWO (2)** ways. (4 marks)
- c) *Hill climbing* is a well-known *local search algorithm*.
- (i) Briefly describe **TWO (2)** advantages of using it. (2 marks)
 - (ii) Briefly describe **TWO (2)** disadvantages of using it. (2 marks)
 - (iii) Provide **TWO (2)** other examples of *local search algorithms*. (2 marks)
- d) A *state-space landscape* is a graphical representation of the states of a problem reachable by a search algorithm, which is usually used to guide the search by visualising how solutions are distributed and how they can be improved.
- (i) *Current state, global maximum, local maximum, flat local maximum, and shoulder* are **FIVE (5)** important elements commonly found in a *state-space landscape*. Briefly describe each of them. (5 marks)
 - (ii) Sketch **ONE (1)** *state-space landscape* that includes the **FIVE (5)** elements stated in **Question 1 d) (i)**, with elevation defined by the values of an *objective function*. You are required to label the X-axis, Y-axis, and all the **FIVE (5)** elements. (8 marks)

[Total: 25 marks]

AMCS2104 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 2**

- a) Most definitions of Artificial Intelligence (AI) can be categorised along two dimensions: *human performance* vs. *rationality* and *thought processes* vs. *behaviour*, resulting in four approaches to AI development. Identify these **FOUR (4)** approaches. (4 marks)
- b) One of the four approaches stated in **Question 2 a)** involves designing and developing agents that perceive their environments through sensors and act upon the environments through actuators. Such agents typically interact with their environments in two ways. Identify the **TWO (2)** types of interaction. Then, differentiate between them. (2 + 2 marks)
- c) Suppose an autonomous driving car named *SmartCar* is an example agent stated in **Question 2 b)**. The problems that *SmartCar* aims to solve can be described using the *PEAS* framework.
- (i) State what *PEAS* stands for. (2 marks)
- (ii) Outline a *PEAS* description for *SmartCar*'s tasks. (4 marks)
- d) Suppose *TAR-GPS* is a GPS system designed by TAR UMT students and installed in *SmartCar* to find the shortest path from Station A to Station H, as shown in **Figure 1**. **Figure 1** illustrates a directed graph of eight stations with edge weights representing distances (km) between stations, while **Table 1** lists the heuristic cost of each station relative to Station H.

**Figure 1.** A map with eight stations.**Table 1:** Heuristic cost of each station.

Station	A	B	C	D	E	F	G	H
Euclidean distance	8	11	9	6	4	2	3	0

- (i) Based on the information provided in **Figure 1** and **Table 1**, illustrate how the *A** search algorithm searches for a path to Station H starting from Station A by drawing **ONE (1)** search tree, assuming nodes are expanded in alphabetical order (if applicable). (8 marks)
- (ii) Identify the *search path* explored and the *solution path* returned. (2 marks)
- (iii) Calculate the *total cost* of the *solution path* returned. (1 mark)

[Total: 25 marks]

Question 3

- a) The commonly studied games in AI usually share certain characteristics. Identify **FIVE (5)** of these characteristics. (5 marks)
- b) Answer the following questions based on the simple game tree provided in **Figure 2**.

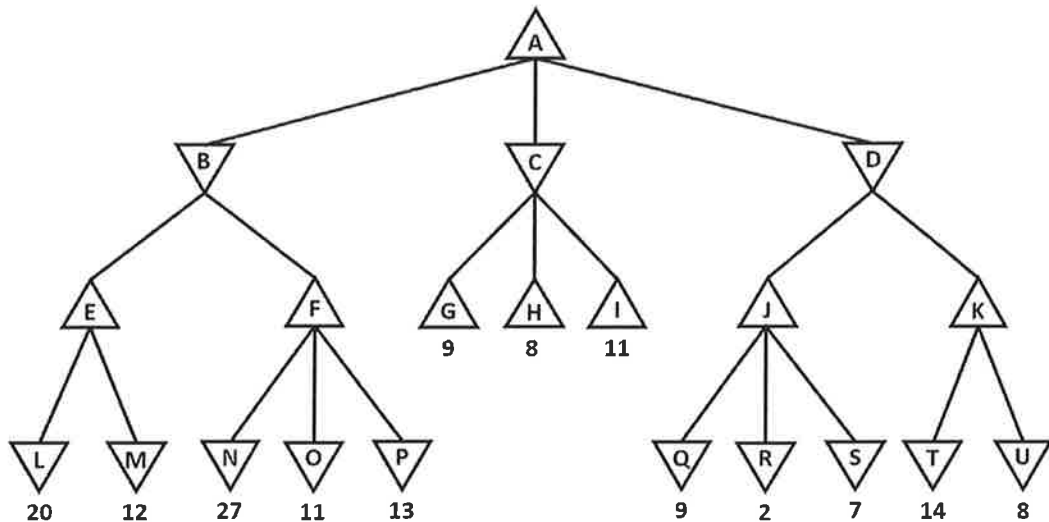


Figure 2. A simple game tree

- (i) Identify the *minimax values* for **EIGHT (8)** nodes: A to F, J and K. (4 marks)
- (ii) Identify the *best move* at the root node. Justify your answer. (1 + 1 marks)
- (iii) Apply minimax search with *alpha-beta pruning* in *left-to-right order* and identify nodes that would be pruned. Then, apply minimax search with *alpha-beta pruning* in *right-to-left order* and identify nodes that would be pruned. For each order, you must demonstrate how *alpha-beta pruning* works on the game tree provided in **Figure 2**. (12 marks)
- (iv) Derive a conclusion based on your answer to **Question 3 b) (iii)**. (2 marks)

[Total: 25 marks]

AMCS2104 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4**

- a) A *Constraint Satisfaction Problem (CSP)* consists of variables, domains, and constraints. Solving a *CSP* therefore involves assigning values to variables while satisfying all specified constraints. There are several types of assignments, including *consistent*, *legal*, *partial*, and *complete* assignments.
- (i) Differentiate between these **FOUR (4)** types of assignments. (4 marks)
- (ii) Define a *solution* to a *CSP*. (2 marks)
- b) Sudoku is a puzzle game that involves the logical placement of numbers and has introduced millions of people to the concept of *CSP*. **Figure 3** shows an example of a 4×4 Sudoku grid consisting of 16 cells, some of which are pre-filled with digits from 2 to 4. The objective is to fill the grid such that each row, column, and 2×2 sub-grid contains each digit—from 1 to 4—exactly once. Answer the following two questions.

	1	2	3	4
A		2	3	4
B	3			2
C	2			3
D			2	

Figure 3. State *x* of the 4×4 Sudoku.

- (i) Suppose the *AllDiff* global constraint is applied to solve the Sudoku puzzle in **Figure 3**. Analyse the Sudoku puzzle as a *CSP* by clearly identifying the *variables*, the *domain* of each variable, and the *constraints*, expressed using the *AllDiff* constraint. (9 marks)
- (ii) To simplify implementation, suppose all *AllDiff* constraints are expanded into binary constraints, allowing the *AC-3* algorithm—an algorithm for enforcing arc consistency—to be applied directly. Then, explain, step by step, how the values of A1, B2, B3, C3, and D4 are determined in sequence, by specifying which constraint is applied at each step and how it leads to the value assignment. (10 marks)

[Total: 25 marks]